Keeping the House in the Family: Non-Market Transfers and Property-Tax Incentives *

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Abstract

A large share of American homes change hands without a market. Using 144 million residential deed events from 2007 to 2023, this paper provides national micro-level evidence on the prevalence and size of intra-family housing transfers. We find 17 family transfers per 100 market sales under a conservative same-surname definition (25 under a wider one), and this family channel is largest in California, where heirs could inherit a parent's low property-tax assessment. California's Proposition 19 provides a natural experiment to study how inherited property-tax assessments shape this channel: transfers rose ahead of the February 2021 deadline, with the largest seasonally adjusted spike in the deadline month (although California transfers were already elevated before the measure reached the ballot), and then declined once it was implemented. Net of California's market sales, a triple difference shows family transfers decreased by 22.5% in the post-reform period, roughly 27,355 missing deeds annually. A simple framework shows that when heirs inherit a below-market property-tax assessment along with the house, as permitted by California's Propositions 13, 58, and 193, families keep homes that the market values more. Using the framework, we estimate the tax wedge and quantify the deadweight loss from the misallocation. Our results suggest that family transfers are not outside the housing market but are one of its allocation mechanisms and property-tax rules can shift scarce homes from price-based matching to lineage-based retention.

JEL classification: R21, R31, H22, D64

Keywords: housing supply, intergenerational transfers, bequests, property taxes, Proposition 19

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1 Introduction

The housing market that economists study is the one with prices. For half a century, that market has been one of the most productive objects in applied economics. Hedonic methods recover what buyers pay for each feature of a home (Rosen, 1974), repeat-sales indexes track how the value of the same house moves over time (Bailey et al., 1963; Case and Shiller, 1989), and search models explain why homes sit unsold, why prices and volume rise together, and why sellers hold out for prices the market will not pay (Genesove and Mayer, 2001; Genesove and Han, 2012; Ngai and Tenreyro, 2014; Piazzesi et al., 2020; Wheaton, 1990). The same transactions tell us how regulation limits new supply (Glaeser et al., 2005; Hsieh and Moretti, 2019) and how often owners move when prices fall or rates rise (Ferreira et al., 2010; Fonseca and Liu, 2024). Housing turnover is measured through market sales. Sales are recorded, indexed and aggregated. Each of these literatures starts, sensibly, from a sale between strangers, because presumably only such a sale reveals what a home is worth. However, many homes change hands without a market. Families transfer homes to each other by gift, inheritance, and other non-market conveyances – a father quitclaiming a bungalow to his son, or siblings dividing a family estate among themselves. These deed transfers are recorded, but they usually do not generate a market price, and thereby ignored by the literature as "non-arms length" transactions. For anyone building a price index, that filter is the right one. A deed from a mother to her daughter for one dollar says little about what the house is worth. But the filter also discards a question: who ends up living in a home when no price is ever consulted? An older literature does study what passes between generations, yet it focuses on dollars rather than houses. Economists have long disagreed about how much of American wealth is inherited rather than saved over a lifetime (Kotlikoff and Summers, 1981; Modigliani, 1988), and about why parents give their wealth to their children: to help the child who earns less, to reward the child who visits, or because they did not expect to die so soon (Becker and Tomes, 1979; Bernheim et al., 1985; Kopczuk, 2007; Kopczuk and Lupton, 2007; McGarry, 1999). Inherited wealth has regained weight in rich economies (Piketty, 2011), the home is the largest asset in the portfolio of the typical middle-class American family (Kuhn et al., 2020), and housing wealth persists strongly across generations (Binder et al., 2026; Charles and Hurst, 2003). Yet the bequest literature treats a house as a sum in an estate, while the housing literature treats the same deed as noise to be cleaned out of a price index. Neither follows the home itself. That omission matters because houses are durable and outlive their owners (Glaeser and Gyourko, 2005), and when prices do not allocate homes, nothing ensures that each home goes to the person who values it most. Rent

control misallocates apartments across tenants (Glaeser and Luttmer, 2003). Property taxes can do the same thing to owners. Under an assessment cap, such as California's Proposition 13, a home is taxed on its purchase price plus at most two percent a year, so a long-time owner's tax bill falls far below what a new buyer would pay for the same house. Selling forfeits that discount, because moving even across the street means paying the market tax on the next home. The low bill is therefore a reward for staying put, and it should lock owners in, as simulations predicted that it would (O'Sullivan et al., 1995). But the first comparison of California's metropolitan areas with those of other states found no clear change in mobility after 1978 (Nagy, 1997). With longer data, the property tax lock-in became visible: California owners now stay in their homes longer than owners in comparable states (Wasi and White, 2005), and letting older Californians carry their assessment to a new home made them more likely to move (Ferreira, 2010). Florida's caps on assessment growth have raised the same question (Ihlanfeldt, 2011). The tax discount works much like a below-market mortgage rate, which also keeps owners from moving (Ferreira et al., 2010; Fonseca and Liu, 2024). Recent work shows how such taxes shape who holds which homes when buyers face financial constraints (Coven et al., 2026). Transaction taxes, for their part, change when and whether homes trade at all (Best and Kleven, 2018; Hilber and Lyytikäinen, 2017; Kopczuk and Munroe, 2015). A family transfer can have both features at once. The home moves without a price, and in California it could, until recently, carry the parent's tax basis with it. Before Proposition 19, the inheritance exclusion let the lock-in outlive the owner. An heir who kept the house kept the parent's tax bill, sometimes decades old. An heir who sold it gave that discount up, because the buyer would be taxed at market value. The older homes that families keep are also the homes that would otherwise filter down to new and often poorer buyers (Rosenthal, 2014). In this paper, we study one of the largest non-market channels of housing turnover: within-family transfers. We document their scale and persistence, and we study how they respond to property-tax incentives. We find that this channel is not small. From 2007 through 2023, county recorders logged 13.6 million family-transfer deeds, about 17 for every 100 market sales. The annual ratio was 0.20 to 0.22 through 2012 and fell to 0.14 by 2022–2023. Much of that fall reflects how we count. Our main definition requires the buyer and seller of an interfamily deed to share a surname, and the share of interfamily deeds without a surname match rose from 17 to 38 percent. Adding those deeds gives about 25 transfers per 100 market sales (Table 2), and that wider ratio is roughly flat. The main count is therefore conservative and, in practice, a lower bound: at most 6.6 percent of same-surname deeds are coded arm's-length, while the 6.1 million

excluded deeds without a surname match likely include many kin with different surnames, such as married daughters. Understanding this channel requires measuring both its scale and the institutions that shape it.

This paper has three parts. First, it establishes national facts about within-family transfers using 144 million residential deed transfer events. We document that family deeds represent a substantial flow alongside market sales in every year. Their ratio to market sales is highest in California (36 per 100, about twice the national ratio), where the tax code historically allowed heirs to inherit a property's assessment as well as the property. Their share rises across middle-aged housing vintages, although it is not monotone over all ages. Many properties remain unsold long after a family transfer.

Second, the paper examines how tax incentives shape recorded family transfers using California's Proposition 19. Under Proposition 13, Propositions 58 and 193, growth rate of assessment values of homes in California were capped at 2 percent per year, and eligible intergenerational transfers could preserve the parent's assessed value, thereby allowing heirs to inherit a below-market property-tax assessment along with the house. Proposition 19 narrowed this exclusion in February 2021, retaining conditional protection for a principal residence occupied by the heir and for family farms while removing it for other qualifying transfers (California State Board of Equalization, 2025). We use the ballot placement and implementation of Proposition 19 as a natural experiment to study how inherited property-tax assessments shape the family transfer channel. We find that California family deeds ran 21.4 percent above a simple counterfactual from the June 2020 ballot placement to the February 2021 deadline, and a monthly event study shows the largest seasonally adjusted spike in the deadline month. That 21.4 percent overstates anticipation. California deeds were already above the counterfactual in early 2020, before the ballot placement, partly a rebound from an unusually low 2018–2019 base. Family deeds then fell to a persistently lower level. A triple difference that compares family deeds with market sales, in California and in other states, before and after the reform, shows family transfers 22.5 percent lower in 2022–2023 net of California's market-sale decline, roughly 27,355 missing deeds annually when scaled by the 2018–2019 baseline. A synthetic control and a monthly event study trace the same persistent post-reform gap.

Third, we develop a simple framework that quantifies the potential allocation gains from removing the inherited-assessment advantage. The framework follows a marginal heir's decision to keep a home or sell it, and the components involved in that decision: the value of the home to the heir, the market value of the home, and the inherited property-tax assessment. The framework shows that the tax advantage tips only those heirs who

value the home less than the market does, and we measure the distortion using the tax wedge for each parcel. We estimate each component of the tax wedge using the 2018–2019 cohort of California family transfers, and we combine the estimated tax wedge with our framework's predictions to quantify the potential deadweight loss from the misallocation. The homes that California families passed down in 2018–2019 carried a median wedge of $2,057 a year in 2023 dollars, including homes with zero wedges. Their total inherited tax savings were about $249 million a year, a transfer from other taxpayers to heirs rather than a loss to the economy. The deadweight loss is the much smaller triangle beneath that transfer: if each of the 27,355 missing deeds a year sent a home to a higher-value user, assigning each home the median wedge and a distortion equal to half that wedge would yield about $28.1 million a year of avoided deadweight loss per annual cohort.

The paper connects the economics of bequests and inter-vivos transfers (Bernheim et al., 1985; Kopczuk, 2007; McGarry, 1999; Piketty, 2011) to an asset-level record of housing ownership. Research on intergenerational wealth transmission (Charles and Hurst, 2003) motivates the importance of the asset, while evidence on heirs' property documents a distinct source of title-related immobility (Burnett and Winters-Michaud, 2025; Dobbs and Johnson Gaither, 2023). Recent work studies intergenerational housing-capital persistence using linked Census, property, and income-tax records (Binder et al., 2026). Our deed-based object is the conveyance of particular assets, rather than the persistence of children's housing holdings. Closely related work by Baker et al. (2026) uses parent-child exclusion claims in Los Angeles and San Francisco to study Proposition 19 retiming and subsequent sales. The present national measurement and California-wide comparisons complement that evidence. Our recorder-based family proxy is not the population of confirmed parent-child exclusion claims: it includes spouses, who are family but not the transfers Proposition 19 targets, and it misses heirs with a different surname. The nearest descriptive antecedents also include industry tabulations (Delventhal, 2026) and the LAO's assessment-roll accounting.

The analysis also extends the property-tax and housing lock-in literature (Coven et al., 2026; Ferreira, 2010; Fonseca and Liu, 2024; Wasi and White, 2005) to intergenerational title transfers. The transaction-tax literature (Best and Kleven, 2018; Hilber and Lyytikäinen, 2017; Kopczuk and Munroe, 2015; Slemrod et al., 2017) provides a comparison for timing and volume responses, although a deadline response alone does not reveal the valuation gaps required for welfare. Misallocation concerns, as in Glaeser and Luttmer (2003), require evidence or assumptions about forgone surplus. Older housing can serve an important filtering role (Rosenthal, 2014), but a family-owned rental may

already provide those services to a tenant.

Section 2 describes the data and classification. Section 3 presents the four national facts. Section 4 studies the policy response, and Section 5 develops the framework and conditional welfare calibration. Section 6 concludes.

2 Data and the Anatomy of a Non-Market Transfer

2.1 Deeds

Property records come from CoreLogic's Owner Transfer file (CoreLogic, 2024), obtained through an academic license, which compiles the universe of recorded deeds from county recorder offices. A deed is filed whenever the legal ownership of a parcel changes—not only when a house is sold. The same administrative trail that records a $400,000 sale records a mother quitclaiming her house to her son for nothing. Market-based datasets (MLS listings, repeat-sales indices, mortgage originations) discard these zero-price events by construction. Here they are the object of study.

The extract covers all fifty states and the District of Columbia. Coverage is dense from 2007 through mid-2024; we use 2007–2023 for annual statistics and treat 2024 as a partial year. Restricting to residential parcels and deduplicating to one event per parcel per derived sale date yields 148.7 million deed events (144.0 million in 2007–2023), linked through a permanent parcel identifier that follows each house across successive owners.

2.2 Classifying transfers

CoreLogic flags each record with a primary category—arm's length or not—and an interfamily indicator. CoreLogic does not document how the indicator is built. In our audit of the file it behaves like a simple rule: it is set on nearly every deed whose buyer and seller share a surname, and on family- or death-related deed types that record no consideration. It does not verify a relationship. We sharpen these flags into a taxonomy using the buyer and seller names on the deed:

1. **Market sale**: arm's-length category, not flagged interfamily, with no price floor because non-disclosure states withhold recorded prices on many genuine sales (78.5 million events, 2007–2023, of which 8.7 percent record a zero or missing price).

2. **Same-person retitle**: the named principal is the same person on both sides of the deed—an owner adding or removing a co-owner, potentially at marriage, divorce,

or refinancing (7.1 million). The matched name remains, but ownership shares can change. These deeds are tracked separately and excluded from the family definition.

3. **Family transfer, same surname**: flagged interfamily, buyer and seller sharing a surname but not the same person, no trust detected (13.6 million). Together with a separately coded estate/executor residual (1,905 deeds), this class is our main family definition, used throughout the main analysis.

4. **Interfamily, no surname match**: flagged interfamily without a detected surname match after the earlier exclusions (6.1 million). Missing surname fields and transfers involving entities can enter this group. Because the relationship cannot be verified, it is excluded from the main family definition and enters only the wide definition in Table 2.

5. **Interfamily trust deed**: the interfamily flag is set and a primary party name contains a trust keyword (12.1 million). We split these by name. A trust that shares a surname with the other party but names a different individual marks a trust-mediated family transfer (1.19 million, a lower bound); adding trust-to-person deeds whose trust name carries no given name gives an upper bound of 1.61 million. The other 10.5 million are a person moving property into or out of the person's own trust, or trust deeds with no surname link. Neither group enters the main family definition; Table 2 reports the trust-mediated transfers as a separate range. The split rests on names alone and does not verify control.

6. **Other non-arm's-length**: the remainder—divorce divisions across surnames, title corrections, nominal-price conveyances, and foreclosure-related instruments (26.6 million).

The main definition is a conservative count and, in practice, a lower bound on family transfers. A shared surname can be a coincidence, but at most 6.6 percent of same-surname deeds are coded arm's length (4.4 percent at prices of $100,000 or more). That error is small next to the 6.1 million flagged deeds without a surname match that the definition leaves out, many of which are likely relatives with different surnames, such as married daughters. The count also includes transfers between spouses, who are family but are not the parent-to-child transfers that Proposition 19 targets.

Table 1 reports observable differences across the classes. Same-surname family deeds differ sharply from the defined market-sales class: 90 percent of same-surname transfers record a zero or missing price, against 9 percent of market sales (mostly unpriced sales

Table 1: Residential deed events by class, 2007–2023

Event class	N (2007–2023)	Zero/no price (%)	Median price[a]	Quitclaim share (%)	Absentee recipient (%)[b]
Market sale (arm's length)	78,470,563	8.7	$221k	2.9	28.5
Other non-arm's-length	26,588,068	57.1	$64k	19.4	58.1
Family transfer: same surname	13,610,459	90.3	$97k	39.6	17.8
Interfamily trust deed	12,074,757	96.1	$130k	30.0	25.6
Same-person retitle (co-owner change)	7,123,579	92.3	$63k	40.3	16.9
Interfamily, no surname match	6,073,863	88.3	$155k	26.4	34.8
Estate deed (non-interfamily)	56,056	7.2	$167k	1.9	91.1
Family transfer: estate/executor	1,905	82.9	$596k	8.6	35.6

Notes: Authors' classification of CoreLogic Owner Transfer records, residential parcels, deduplicated to one event per parcel-date. Market sales are arm's-length (category A), non-interfamily deeds with no price floor, so they include deeds recorded without a price (common in non-disclosure states). The main family definition is same surname plus estate/executor; interfamily deeds without a surname match enter only the wide definition in Table 2. Interfamily trust deeds are mostly self-transfers or trust deeds with no surname link; Table 2 separates the trust-mediated family transfers. [a]Median of strictly positive recorded prices. [b]Share of events whose buyer mailing ZIP differs from the property ZIP, among events where both are observed.

in non-disclosure states) and 57 percent of the residual non-arm's-length class. A missing price alone therefore does not separate a family transfer from a sale. Forty percent of same-surname transfers travel by quitclaim deed—the instrument of gift—against 3 percent of market sales. Where a price is recorded at all in a same-surname transfer, the median is $97,000, well under half the $221,000 median market price: the comparison is descriptive because the houses and transferred interests can differ. Corporate buyers, who take 12 percent of market purchases, are essentially absent from same-surname transfers. These patterns are consistent with gifts and succession, but do not verify their prevalence or exclude other transactions. One note on probate: formal estate keywords ("estate of," executor, administrator) rarely appear in the recorded name fields, and the earlier trust, same-person, and surname rules also take precedence over the estate rule. The small residual estate class (1,905 interfamily deeds) therefore cannot measure all successions; another 56,056 estate-keyword deeds lack the interfamily flag and stay out of the family definition.

3 Within-Family Housing Transfers: National Facts

Fact 1: For every 100 market sales, 17 family-transfer deeds are recorded. Table 2 reports annual volumes and Figure 1 shows the pooled scale and annual path of the family channel. Between 0.65 and 1.05 million family-transfer deeds are recorded

every year (same-surname and estate/executor), against 3.4 to 6.4 million market sales: a family:market ratio between 0.14 and 0.22 in every year of the sample, with a 2007–2023 aggregate of 0.17, i.e. 17 family-transfer deeds per 100 market sales. The channel is not a recession artifact. The ratio stays between 0.20 and 0.22 from 2007 to 2012, while market sales were depressed after the 2008–2009 recession, and then declines to 0.14 by 2022–2023. Most of that decline is definitional: the share of interfamily deeds without a surname match, which the main definition excludes, rises from 17 percent in 2007 to 38 percent in 2023. Adding those 6.1 million deeds back gives 19.7 million family deeds, 25 per 100 market sales, and a ratio that ends where it starts (0.24 in 2007, 0.23 in 2023). The main count is therefore conservative and, in practice, a lower bound. At most 6.6 percent of same-surname deeds are coded arm's-length (4.4 percent at $100,000 or more), which bounds coincidental surname matches; that is small next to the 6.1 million excluded deeds, many of which are likely transfers to kin with a different surname, such as married daughters. The count also includes spouses, who are family but not the parent-child transfers Proposition 19 targets. It excludes the 1.2 to 1.6 million trust-mediated family transfers (1.5 to 2.1 per 100 market sales), which more than double over the sample (from 43,000 in 2007 to 98,000 in 2023, lower bound); adding them raises the aggregate ratio to 0.19. It also excludes 10.5 million trust deeds that are self-transfers or have no surname link, 7.1 million same-person retitles, and 26.6 million other non-market conveyances.[1]

Fact 2: Family transfers are concentrated in California. Figure 2 ranks states by the family:market ratio over 2007–2023. California leads the nation at 0.36—36 family transfers for every 100 market sales, 2.1 times the pooled national ratio of 0.17. Utah is second at 0.29. California is exactly where acquisition-value assessment made the inherited tax basis most valuable, and it also accounts for 39 percent of all interfamily trust deeds in the country (4.8 million over the sample period, nearly twice its family transfers), consistent with extensive estate-planning activity. Cross-state correlations are not causal evidence—demography, land tenure, and recording practice all differ—but the cross-section puts the tax-subsidy hypothesis on trial, and Section 4 tests it for California.

Fact 3: Family transfer shares are higher across middle-aged housing vintages. Linking deed events to property characteristics (60 percent of 2017–2023 selected family/trust/market events match the available property-characteristics snapshot;

[1]Restricting the denominator to existing homes (excluding new-construction first sales) raises the aggregate family:market ratio to 0.18. On a per-property rather than per-deed basis, the 13.6 million family-transfer deeds cover 11.8 million distinct parcels (1.15 deeds per parcel).

Table 2: The parallel housing market: annual volumes under alternative family definitions

Year	Market sales	Same surname	Same surname + estate (main)	Same surname + estate + other[a]	Family trust transfers[b]	Trust self or unlinked[c]	Family:market ratio (main)
2007	4,431,630	884,266	884,326	1,068,273	43,128–72,294	446,456	0.20
2008	3,761,060	763,923	764,010	940,843	44,863–69,137	408,452	0.20
2009	3,681,488	764,893	764,908	954,929	49,186–81,668	431,772	0.21
2010	3,484,935	723,862	723,897	959,657	54,544–85,625	506,463	0.21
2011	3,394,405	726,964	727,000	977,935	57,961–86,523	531,198	0.21
2012	3,787,240	827,415	827,593	1,136,675	67,508–102,145	629,607	0.22
2013	4,171,219	803,373	803,409	1,132,061	65,807–93,212	587,433	0.19
2014	4,249,438	718,340	718,369	1,036,746	65,284–87,986	565,440	0.17
2015	4,704,221	778,399	778,430	1,130,271	70,368–94,960	617,894	0.17
2016	4,931,194	804,689	804,750	1,178,714	74,211–98,569	640,914	0.16
2017	5,100,699	772,432	772,481	1,188,626	72,154–93,986	606,541	0.15
2018	5,252,021	783,314	783,422	1,258,262	74,828–94,013	593,562	0.15
2019	5,395,160	819,739	819,830	1,315,281	76,281–94,385	647,141	0.15
2020	5,601,162	946,282	946,477	1,439,289	86,781–109,438	778,883	0.17
2021	6,376,852	1,047,039	1,047,335	1,622,749	99,060–123,903	842,246	0.16
2022	5,587,888	792,479	792,991	1,284,736	94,006–111,034	790,604	0.14
2023	4,559,951	653,050	653,136	1,061,180	97,812–113,959	837,314	0.14

Notes: Residential deed events. The main family definition ("same surname + estate") counts interfamily-flagged deeds with a buyer–seller surname match plus estate/executor deeds, and is used in all later analysis. [a]Adds interfamily-flagged deeds without a surname match. [b]Trust-mediated family transfers, lower–upper bound: lower = trust deeds whose trust shares a surname with the counterparty but names a different individual; upper adds trust-to-person deeds whose trust name carries no given name. [c]Interfamily-flagged trust deeds classified as a person moving property to or from the person's own trust, plus those with no surname link; excluded from every family column. Ratio = main family definition / market sales.

Appendix A), the family share of within-class traffic rises with structure age: family transfers are 13 percent of combined family-plus-market events for houses of ten to thirty years, 14 percent at thirty to fifty years, and 16 percent at fifty to seventy-five years, before falling to 14 percent for homes aged 75 or more. The gradient is therefore not monotone over all vintages; the under-ten-year bin also has very few matched events. The family share is lower on parcels with senior tax exemptions (13 versus 15 percent)—an association measured with a historical snapshot, and one that reverses sign when market sales must carry a recorded price of at least $10,000. Across the within-state value distribution the family share is roughly flat at 15 percent through the bottom four quintiles and falls to 13 percent in the top. The houses that families keep lean toward middle-aged stock and away from the most expensive homes. That is the stock through which housing filters down to lower-income buyers (Rosenthal, 2014).

Family deeds are less common in condominiums and townhouses. Using recorded land-use codes, family deeds account for 16.0 percent of combined family-plus-market deeds for single-family properties, 15.8 percent for duplexes, 9.1 percent for condominiums, and 8.5 percent for townhouses or rowhouses. On common state-year support, weighting

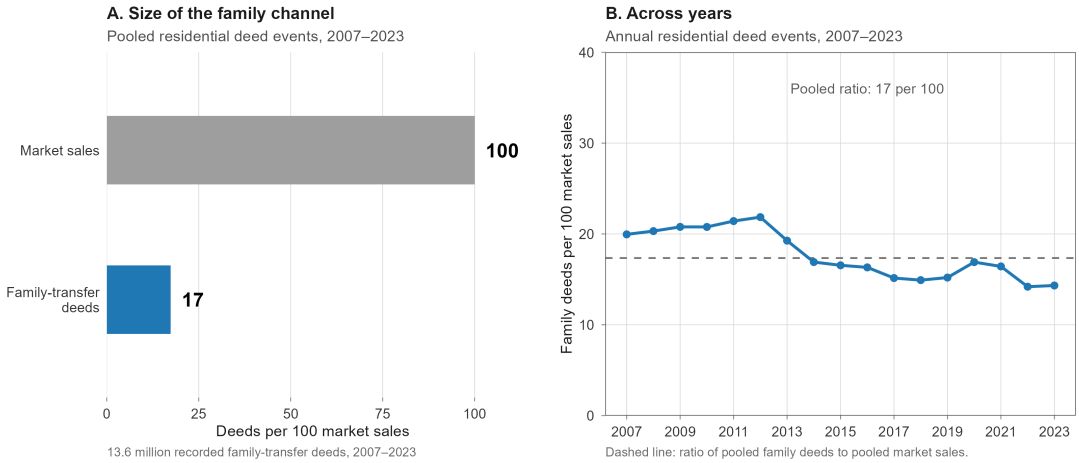


Figure 1: Size and annual path of the family-transfer channel, 2007–2023. Panel A normalizes pooled market sales to 100 and shows approximately 17 family-transfer deeds per 100 market sales. Panel B reports annual family-transfer deeds per 100 market sales; the dashed line is the pooled ratio (17.3). Family transfers combine the same-surname and estate/executor classes (the main definition); they exclude interfamily deeds without a surname match, trust self-transfers, and trust-mediated transfers. Market sales carry no price floor. Counts refer to deed events, not distinct properties.

each type to the single-family distribution leaves the duplex share 2.2 percentage points lower and the condominium and townhouse shares 7.6 and 6.4 points lower, respectively. Appendix A.1 gives the definitions and denominators. These are static recorded types, not measures of occupancy or rental use; the apartment sample is selectively covered. Duplexes, a common form of small income property, are not transferred within families more often than single-family homes. We do not treat building type as evidence that families pass on housing capital rather than a residence.

Fact 4: Three in five family-transferred homes are still unsold a decade later. What happens to a house after it enters the family channel? Because the parcel identifier links successive deeds, every transfer event can be followed to the property's next arm's-length market sale, if any. We take all events in the 2008–2018 cohorts—guaranteeing at least 66 months of follow-up before the June 2024 censoring date—and estimate Kaplan–Meier survival to first market sale, separately by event class: 8.5 million same-surname family transfers (plus 665 estate/executor deeds), 7.1 million interfamily trust deeds, 4.1 million same-person retitles, and 46.5 million market purchases as the benchmark.

Table 3 and Figure 3 show that family homes lock in. Same-surname transfers start slightly faster than market purchases (7.5 versus 6.7 percent sold within twelve months,

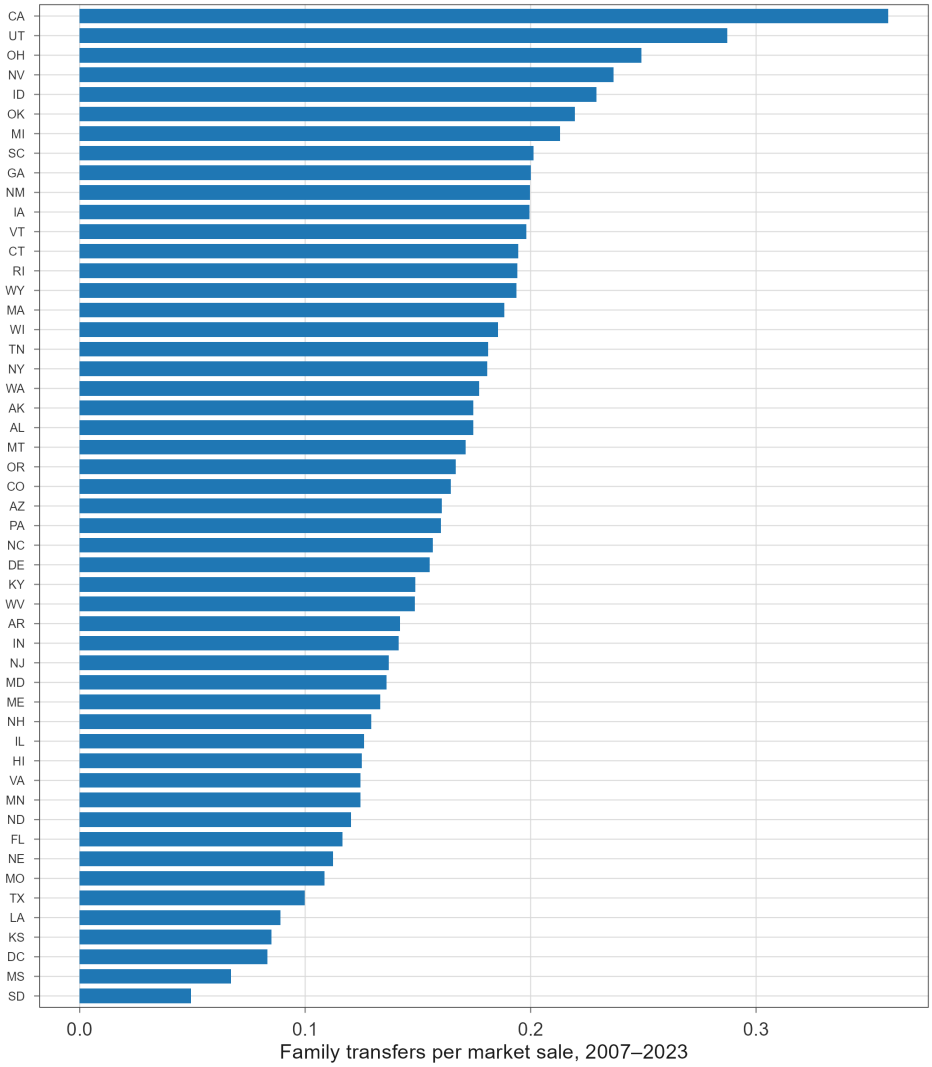


Figure 2: Family transfers per market sale by state, 2007–2023.

12.0 versus 10.9 percent within two years) but fall behind by year five (24.6 versus 26.1 percent) and keep falling behind: by year ten, 40 percent of same-surname family homes have returned to the market against 45 percent of market purchases. The survival curves cross at about month 44, and the five-year family sale rate is below the market rate in each of the eleven annual cohorts from 2008 to 2018. Conditional on surviving the first two years, 31 percent of family homes sell over the following eight years versus 38 percent of market purchases. The market-purchase benchmark includes investors and flippers. Without a price floor it also includes 4.5 million deeds recorded with no price or a price below $10,000, which resell within twelve months more than three times as often as priced sales and so steepen the early market curve. Three in five (60 percent) of the

Table 3: Time to next open-market sale, 2008–2018 event cohorts

		Sold on open market within (%)			
Event class	N	12m	24m	36m	60m
Market sale (arm's length)	46,517,920	6.7	10.9	15.9	26.1
Family transfer: same surname	8,467,604	7.5	12.0	16.5	24.6
Family transfer: estate/executor	665	10.8	15.5	42.4	47.5
Interfamily trust deed	7,107,100	4.9	9.4	13.7	21.9

Notes: Share of events followed by an arm's-length market sale of the same parcel within the stated horizon. All cohorts observed at least 66 months before the June 2024 censoring date.

same-surname homes that passed between relatives in 2008–2018 had not sold on the open market a decade later. An internal descriptive check is that same-person retitles—deeds retaining at least one matched name—track interfamily trust deeds closely (9.7 versus 9.4 percent sold within two years), consistent with both classes often recording paperwork rather than a change in the property's use.

This is a descriptive fact, and selection is its content: families keep the homes they receive. Four caveats discipline its reading. First, the comparison pools states and cohorts as family transfers concentrate in older homes and in low-turnover states, so part of the gap is composition. Second, the two clocks measure different objects: a market purchase starts a fresh tenure, while a family deed often changes title without changing occupancy—a force that should, if anything, *raise* the family class's sale hazard, if family deeds cluster near succession events. Third, some family homes cannot sell for title reasons (heirs' property), a lock of a different kind (Dobbs and Johnson Gaither, 2023). Fourth, interfamily trust deeds —the trust-keyword class, mostly self-transfers—resell even more slowly than same-surname family homes (21.9 versus 24.6 percent sold within five years), a reminder that long-tenure households are exactly the ones that do estate paperwork. These comparisons do not distinguish attachment, rental use, title constraints, and tax incentives. Proposition 19 identifies a response of deeds to a policy change; additional assumptions are needed to connect that response to allocation.

4 Tax Incentives & Family Transfers: Proposition 19

4.1 Institutional background

Proposition 13 (1978) fixed each California parcel's assessed value at its acquisition price, with growth capped at two percent per year; reassessment to market value occurs only

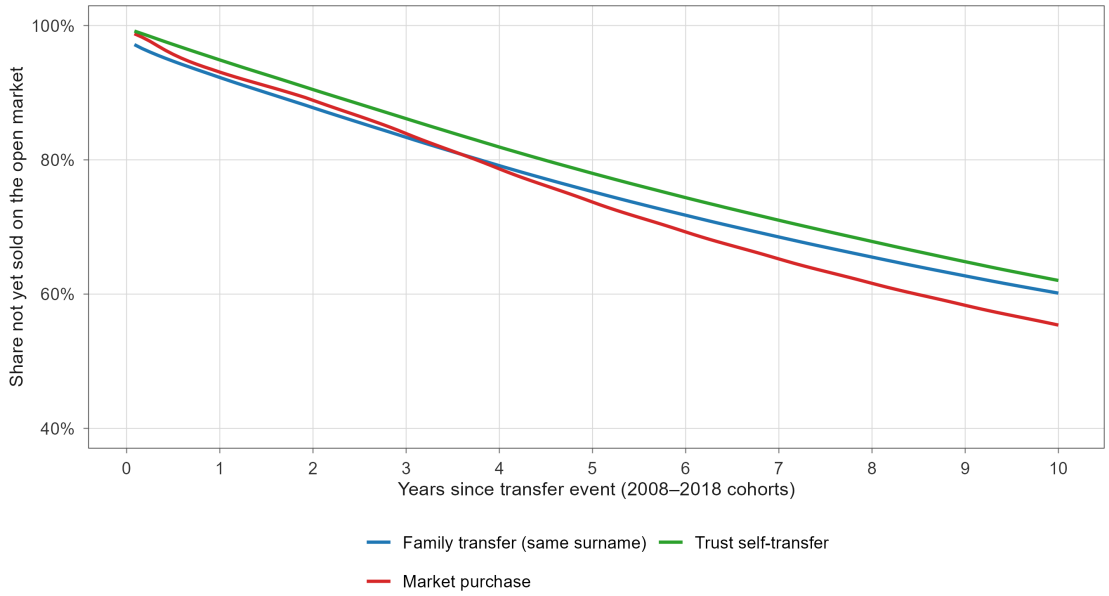


Figure 3: Kaplan–Meier survival to first open-market sale, 2008–2018 event cohorts, censored June 2024. Family transfers are same-surname deeds. Market purchases carry no price floor and include arm's-length deeds with no recorded price or a price below $10,000. The estate/executor class (n=665) is omitted.

at a change in ownership. Proposition 58 (1986) exempted parent-to-child transfers from this reassessment—for a principal residence of unlimited value plus up to $1 million of assessed value of other property—and Proposition 193 (1996) extended the exemption to grandparent-grandchild transfers. A child could therefore inherit not just a house but its decades-old tax bill, and keep that tax bill while renting the house out. The Legislative Analyst's Office estimated that 60,000 to 80,000 inherited properties escaped reassessment each year under this regime (California Legislative Analyst's Office, 2017).

Proposition 19, passed by voters on November 3, 2020, narrowed the exclusion sharply, effective February 16, 2021: it now applies only when the home was the parent's principal residence *and* the child makes it their own principal residence within a year, and only up to the inherited taxable value plus $1 million (inflation-adjusted); the exclusion for rentals, vacation homes, and other non-primary-residence property was eliminated, with a carve-out for family farms (California Legislative Analyst's Office, 2020; California State Board of Equalization, 2025). For an otherwise eligible non-occupant heir, the reform removes the inherited-assessment advantage. This raises the tax cost of retention relative to sale, although the family may respond by changing paperwork, timing, occupancy, or ownership rather than by listing the home. County assessors reported a flood of parent-

child filings ahead of the deadline.[2] The reform's other arm—expanded portability of assessments for movers aged 55 and over, effective April 1, 2021—affects market sales rather than family transfers; market sales therefore need not be an untreated comparison group. The triple difference below uses them as the comparison deed group; the synthetic control does not use them.

Throughout this section, "family transfers" means the main definition (same-surname interfamily deeds plus estate/executor deeds), measured monthly by state from the event panel. Interfamily deeds without a surname match and trust deeds are excluded. Appendix B adds trust deeds to the family count: adding trust distributions gives a similar decline, while adding all trust deeds erases it. The count includes transfers between spouses, which are family transfers but not the parent-child transfers Proposition 19 targets.

Figure 4 plots monthly California family transfers against a transparent counterfactual: California's 2019 volume in the same calendar month, scaled by contemporaneous family-transfer growth in the other 49 states and DC. The counterfactual thus inherits both California's seasonality and national shocks—COVID disruption, the 2020–2021 housing boom, the estate-planning surge—while excluding California from the donor pool. Before the measure reached the ballot, the fit is only moderate: over 2018m1–2020m5, the mean absolute monthly error is 7.8 percent and the RMSPE is 11.4 percent, close to the median state's fit.

California runs above the counterfactual before the measure reached the ballot: 18.5 percent above it over January–May 2020, before the June 24, 2020 ballot placement. The gap does not widen right after the placement: over June–December 2020 California runs 14 percent above the counterfactual. It widens sharply at the deadline, to 52 percent above the counterfactual in January and February 2021. From June 2020 through February 2021, actual family transfers total 142,320 against a counterfactual of 117,231—an excess of 25,089 deeds, 21.4 percent above counterfactual over the whole anticipation period. This figure overstates anticipation. California's 2018–2019 base is unusually low relative to the rest of the country: the log gap between California and other-state family transfers was -1.30 to -1.42 in 2010–2016 but -1.76 and -1.69 in 2018 and 2019, so part of the excess is a rebound from a depressed base. February 2021, ordinarily one of the year's quietest months, ran 57 percent above its counterfactual, the largest monthly excess

[2]Sonoma County, for example, received roughly five times its typical volume of parent-child exclusion filings in the weeks before the February 15, 2021 deadline (North Bay Business Journal, February 2021); the Los Angeles County assessor accumulated a multi-year backlog of roughly 20,000 claims (Los Angeles County Auditor-Controller, 2025).

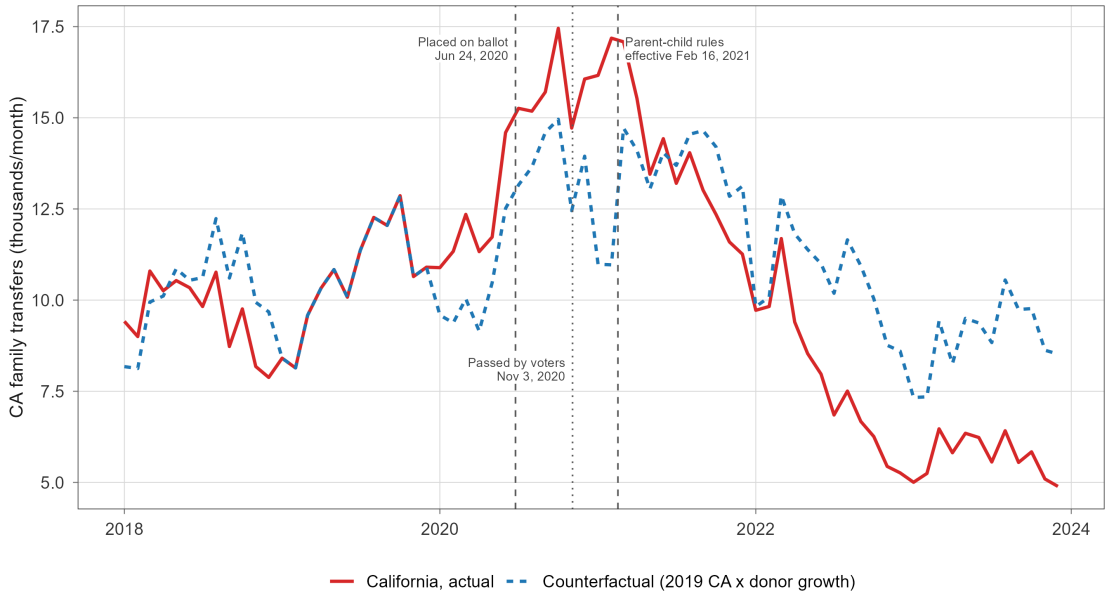


Figure 4: California monthly family transfers (main definition: same-surname plus estate/executor deeds) versus counterfactual (CA 2019 same-calendar-month volume × donor-state growth). Dashed lines: June 24, 2020 (placed on the ballot) and February 16, 2021 (parent-child provisions effective); dotted line: November 3, 2020 (passed by voters).

in the 2018–2023 California series. The spike at the deadline resembles the anticipatory bunching that notch designs recover for taxed market transactions (Best and Kleven, 2018; Slemrod et al., 2017), here in transfers with no price, and is consistent with families retiming successions to keep a tax basis. The concentration around the deadline supports a tax-timing interpretation, although monthly counts do not identify the exact legal transfer date, and the rise before the ballot means this counterfactual cannot cleanly separate the rush from other California-specific growth in 2020.

4.2 Empirical Strategy

We compare California with the other 49 states and the District of Columbia. Identification requires that, absent the reform, California's family-transfer trajectory would have followed the relevant control trajectory.

4.2.1 Synthetic control

Following Abadie et al. (2010), we construct a convex combination of donor states using monthly family-transfer counts normalized by each state's 2018–2019 monthly mean. We

choose weights to match the January 2018–May 2020 trajectory, ending the fitting window before the June 2020 ballot announcement. The post-reform effect is the average proportional gap relative to synthetic California in 2022–2023. Inference uses in-space placebos ranked by the post/pre mean squared prediction error ratio and intervals based on donor-state gaps.

4.2.2 Triple difference

We stack family transfers and market sales and estimate the following triple-difference specification using annual data:

$$\log N_{gsy} = \alpha_{gs} + \lambda_{gy} + \gamma\, \mathbf{1}\{s = \text{CA}\}\mathbf{1}\{y \geq 2022\} + \delta\, \mathbf{1}\{s = \text{CA}\}\mathbf{1}\{y \geq 2022\}\mathbf{1}\{g = F\} + \varepsilon_{gsy}, \tag{1}$$

where N_{gsy} counts deeds of group $g \in \{F, M\}$ (family transfers or market sales) in state s and year y, with state and year fixed effects that vary by group. The coefficient γ is California's post-reform change in market sales, and δ is the triple difference: the change in California family transfers net of the change in California market sales, each relative to other states. Netting market sales matters because California's market sales also fell after 2021, and the reform's portability provisions may affect them. With group-specific effects, δ equals the difference between separate family and market difference-in-differences coefficients. The sample includes 2017–2019 and 2022–2023, excluding the anticipation and transition years 2020–2021. Inference reassigns treatment across states and ranks California's absolute coefficient in the resulting placebo distribution.

4.2.3 Event-study design

To separate anticipation from the post-deadline response, we estimate a monthly event study over 2018–2023 using the following specification:

$$\log F_{st} = \alpha_s + \lambda_t + \sum_{\tau \in \mathcal{T} \setminus \mathcal{B}} \beta_\tau\, \mathbf{1}\{s = \text{CA}\}\mathbf{1}\{t = \tau\} + \varepsilon_{st}, \tag{2}$$

where t indexes calendar months, $\mathcal{T}$ is the sample period, and $\mathcal{B}$ pools January 2018–December 2019 as the omitted baseline. State and year-month fixed effects absorb persistent state differences and common shocks; β_τ measures California's differential change relative to the baseline. We report clustered confidence intervals at the state level.

4.3 Results: The Rush and the post-reform decline

Retiming alone would imply a transitory post-deadline trough followed by recovery. After the deadline, California family transfers fall below the counterfactual by mid-2021 and stay there through 2023, while family transfers in the other states fell back to about their 2019 level in 2022. Retiming cannot be ruled out for 2021–2022: the 25,089 excess anticipation deeds equal about 72 percent of the 35,045-deed shortfall over March 2021–December 2022. The decline that persists through 2022–2023 is the robust result. A donor-weighted synthetic control sharpens the comparison. Following Abadie et al. (2010), we build a synthetic California from a convex combination of the other 49 states and the District of Columbia, with weights chosen to match California's pre-reform family-transfer trajectory. Because California is the largest state—an outlier in the *level* of transfers, which forces a level-matched synthetic onto a single large donor—we match on each state's volume normalized to its own 2018–2019 mean. The synthetic reproduces California's pre-reform path to a root mean squared error of 4.2 percent over January 2018–May 2020 (Figure 5); after the February 2021 deadline, California's family transfers fall back to synthetic California, drop below it from late 2021, and remain below it through 2022–2023. The 2022–2023 level is 29 percent below synthetic California; California's post/pre MSPE ratio ranks second of 51 units (placebo $p = 0.039$).

The triple-difference estimate in Table 4 formalizes this. Using state-by-year counts for 2017–2019 versus 2022–2023[3], family transfers fall by 0.430 log points in California relative to other states (column 1), while market sales fall by 0.175 log points over the same window (column 2), California's general post-rate-shock slump. The triple difference in column 3 nets the two: family transfers fall by 0.255 log points (s.e. 0.030) relative to market sales, a 22.5 percent decline specific to family transfers, or roughly 27,355 transfers per year against the pre-reform baseline. California ranks 6th of 51 in the state-placebo distribution ($p = 0.118$). This estimate implies that, relative to the change in market sales, California had about 22 percent fewer family transfers than it would have absent the reform as seen in the other states.

To distinguish the pre-deadline rush from the subsequent decline, we next turn to monthly event-study evidence. Figure 6 shows monthly event-study results. We regress log family transfers on California-by-month indicators with state and year-month fixed effects, the other 49 states and DC as controls, and the 2018–2019 months as the pooled baseline group. As shown in Figure 6, California is already above its 2018–2019 baseline before the ballot: the January–May 2020 coefficients average about 0.08 log points, and

[3]The transition years 2020–2021 are excluded as contaminated by anticipation and retiming.

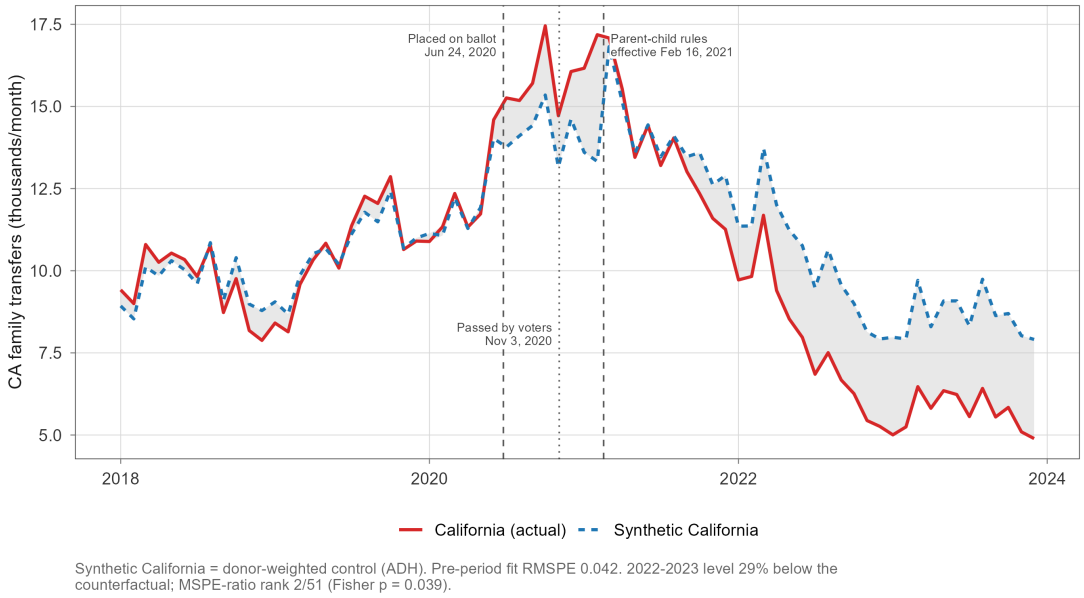


Figure 5: California monthly family transfers against a donor-weighted synthetic California (normalized-index match; Abadie et al., 2010). The series track closely through May 2020; California runs above the synthetic during the anticipation period and below it from late 2021. Dashed lines: placement on the ballot (June 24, 2020) and the parent-child effective date (February 16, 2021); dotted line: passage (November 3, 2020). Weights match January 2018–May 2020.

the February–May coefficients are each significantly positive. The coefficients step up by about 0.12 log points once Proposition 19 reaches the ballot, averaging about 0.20 over June 2020–February 2021. The 2019-scaled counterfactual above shows a smaller and later step, with no widening until the deadline months; the two measures use different baselines (2019 versus pooled 2018–2019), and both put the largest excess at the deadline. The largest seasonally adjusted coefficient is February 2021 (0.33), the deadline month. After the deadline the coefficients fall and settle into a persistent decline, averaging about -0.37 in 2022–2023 and reaching -0.5 by late 2023[4]. The event study shows a step up in anticipation and a post-reform decline; because California was already rising before the ballot, the anticipation evidence is weaker than the decline. One reading is that once the proposition is put on the ballot, the salience of the reform rises, and families rush to transfer properties before the new rules take effect. After the deadline, as the reform is implemented, the number of family transfers drops below the baseline and remains low relative to the baseline, indicating a persistent effect of the reform on family transfers.

Although these results are informative and present us with a clear picture of the ef-

[4]The event study is not net of market sales, the triple difference above provides that comparison.

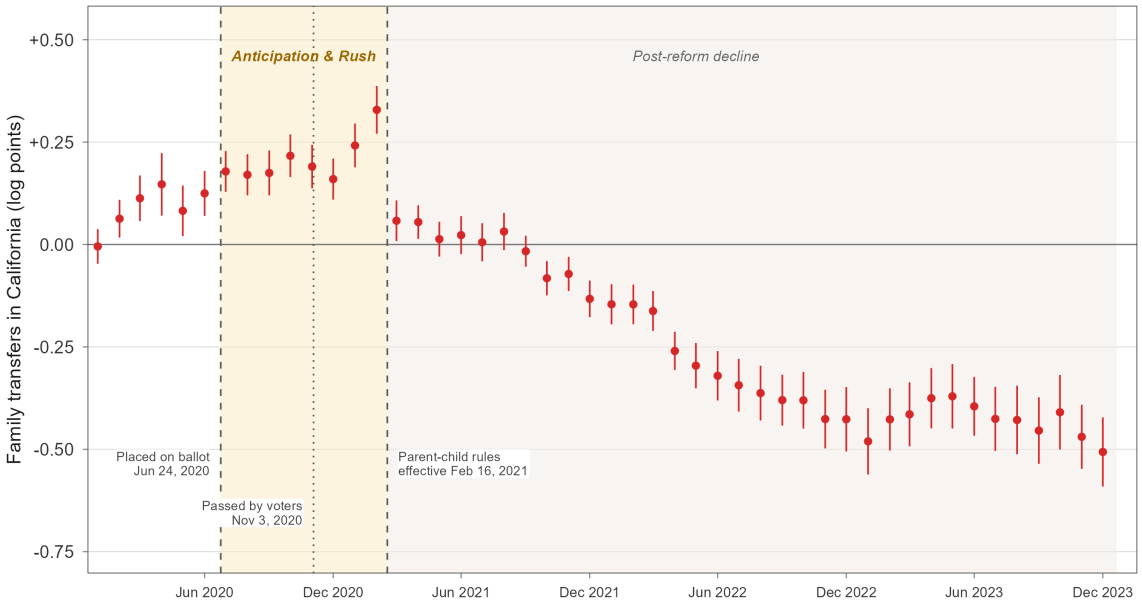


Figure 6: Monthly event study of California family transfers. Points are California-by-month coefficients from a regression of log family transfers on state and year-month fixed effects, with the 2018–2019 months pooled as the omitted baseline and the other 49 states and DC as controls. Whiskers are 95 percent state-clustered confidence intervals. Shaded regions mark the anticipation period, from placement on the ballot (June 24, 2020) to the February 16, 2021 effective date, and the post-reform period. Dashed lines mark the ballot and effective dates; the dotted line marks passage (November 3, 2020).

fect of Proposition 19 on family transfers, we note that the inference is limited to the treated state of California. We have only one treated state, and within that state, all family transfers are affected by the reform. Therefore, these estimates do not have a causal interpretation, as for that, we would need to consider a scenario where we randomize the assignment of the proposition 19 across inherited properties in California. This randomization would allow us to identify the causal effect of the inherited-assessments on family transfers. However, we do not have such a random assignment, and therefore, to better understand the effects of Proposition 19, we need to consider other sources of variation. Next, we develop a simple framework to understand the mechanisms through which Proposition 19 affects family transfers and the potential welfare implications of the reform.

Table 4: Proposition 19: volume, selection, and composition

	Log volume			Sold within 24 months		Absentee recipient
	Family transfers	Market sales	Family vs. market	Family transfers	Market purchases	Family transfers
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
CA × Post × Family			-0.2551			
			(0.0300)			
CA × Post	-0.4300	-0.1749	-0.1749	-0.0167	-0.0111	0.0023
	(0.0335)	(0.0282)	(0.0282)	(0.0037)	(0.0017)	(0.0029)
State-placebo p (DDD)			0.118			
Fixed-effects						
State	Yes	Yes		Yes	Yes	Yes
Year	Yes	Yes				
State-deed group			Yes			
Year-deed group			Yes			
Cohort month				Yes	Yes	Yes
Fit statistics						
Observations	255	255	510	1,836	1,836	1,836
R^2	0.98662	0.98926	0.99303	0.86777	0.94301	0.89863

Clustered (State) standard-errors in parentheses

Notes: Columns (1)-(3): state-year counts, 2017-2019 versus 2022-2023. Column (3) stacks family and market counts with state and year effects that vary by deed group; its CA × Post × Family coefficient is the triple difference, equal to column (1) minus column (2). Columns (4)-(6): transfer-cohort cells, January 2017-December 2018 versus July 2021-June 2022, weighted by cell size. Standard errors are clustered by state and shown for scale; inference with one treated state uses state placebos.

5 Family Channel: A Simple Framework

The policy experiment identifies a response of recorded deeds. Translating that response into welfare requires two further steps: determining which properties change allocation, and valuing those changes. This section makes those steps explicit. Its quantitative results are conditional calibrations of potential gains from the inheritance provisions, not estimates of realized gains or the total welfare effect of Proposition 19.

5.1 Framework

Consider a price-taking marginal market buyer and a marginal heir. Let u_m be the flow per-period value of the house to the marginal market buyer and u_h be the value of the house to the heir, including attachment and the net value of rental use.

Suppose the property-tax rate is t, assessments reset to the sale price when a property is sold, and the discount rate is r. Capitalizing the perpetual tax burden, the marginal market price is

$$p = \frac{u_m}{r + t}. \tag{3}$$

Now consider an heir at the moment of succession who values living in the house, or renting it out, at flow u_h. If the family keeps the house, the heir pays taxes on assessed

value A. If the family sells, the heir receives $\$p$ and the house exits the family channel. Since the heir's inherited assessment is A, keeping the house is optimal when

$$\frac{u_h - tA}{r} \geq p \quad \Longleftrightarrow \quad u_h \geq u_m - t(p - A). \tag{4}$$

Equation (4) contains the framework's key trade-off. If succession triggers reassessment ($A = p$), the condition collapses to $u_h \geq u_m$: the house stays in the family only when the heir genuinely values it more than the market does, and the family channel is allocatively harmless. But if the heir inherits the parent's assessment ($A = A_0 < p$), the family keeps the house whenever $u_h \geq u_m - t(p - A_0)$. Every heir with $u_h \in [u_m - t(p - A_0), u_m)$ retains a house that the market values more than she does. Figure 7 illustrates this interval. The wedge $t(p - A_0)$, the annual tax saving from dynastic holding, is the extra benefit the tax policy pays families to keep the houses in the family channel that would otherwise be sold in the housing market.

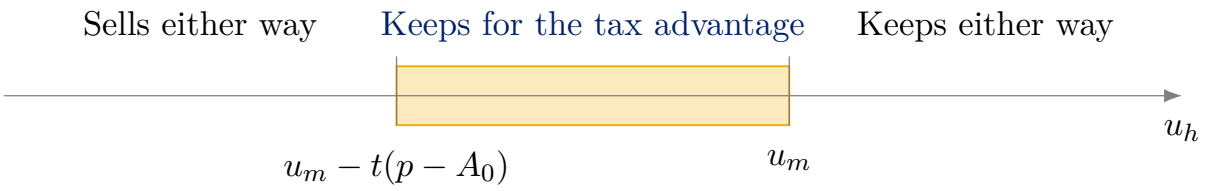


Figure 7: Tax-induced retention. With inherited assessment $A_0 < p$, heirs in the shaded interval $[u_m - t(p - A_0), u_m)$ keep the house despite valuing it less than the market buyer.

In our setup, the succession is a joint decision of the parent and the heir, and the recorded family transfer is the outcome of the keep-or-sell choice. When the family keeps, the county records a family deed; when it cashes out, the family sells directly to a market buyer and no family transfer ever appears[5]. The heir's decision is therefore a function of the tax wedge, the market price, and her own valuation. Thus, making the deed transfer decision endogenous to the tax regime. Introducing a policy reform that removes the inherited assessment ($A_0 \to p$), the following three predictions follow from the framework. First, retiming: because a family transfer must be completed before the deadline to preserve the tax advantage, the families rush to lock in A_0. Families with latent transfer plans and valuation within $[u_m - t(p - A_0), u_m)$ rush to record deeds before the effective date, and recorded transfers collapse afterward (cf. the transaction-tax notch literature:

[5]This simple framework abstracts from intermediary ownership structures and multiple deeds for the same property. An extension could allow a family to transfer the property to a trust before it reaches the market, or could model successive recorded deeds within a single allocation change.

Best and Kleven, 2018; Kopczuk and Munroe, 2015; Slemrod et al., 2017). Second, the extensive margin: with the wedge removed, every heir still facing the succession decision and with valuations in $[u_m - t(p - A_0), u_m)$ switches out of the recorded family-transfer channel. The steady-state volume of family transfers falls by the mass of heirs remaining in the wedge, and the reform potentially releases the homes to the market. Third, selection: the family transfers that survive are those with $u_h \geq u_m$—heirs who genuinely want the house, disproportionately owner-occupants, whom the surviving exclusion still favors. The framework therefore predicts that post-reform family transfers should show a lower subsequent market-sale hazard and a lower absentee-recipient share—not because the reform failed, but because the marginal would-have-been-landlord heirs no longer take the title. The data do not confirm this third prediction. In Table 4, the absentee-recipient share of California family transfers does not fall (column 6: +0.2 percentage points), and their 24-month resale rate falls only slightly more than that of market purchases (1.7 versus 1.1 percentage points, columns 4 and 5).

5.2 Estimating the tax wedge

In this section, we estimate the tax wedge $w = t(p - A_0)$ for California's homes that inherit the parent's assessment explained in Section 5. Since the wedge is not recorded in any deed, we build it parcel by parcel from three components: the tax rate t, the inherited assessment A_0, and the market value p. Each term is measured at a January 1 lien date, the date on which California fixes assessments. A deed transfer associated with market sale records a price, and family deeds usually record none. CoreLogic's assessment roll also stops in 2008–2009. We therefore build each of the three terms parcel by parcel and estimate

$$\hat{w}_i = \hat{t}_i \max\big(\hat{p}_i - \hat{A}_i,\, 0\big) \tag{5}$$

for the 7,852,730 single-family and condominium parcels on the 2008–2009 California roll. The main lien date is January 2019, the year before Proposition 19 is introduced on the ballot. Further, January 2017 is used for homes inherited in 2018–2019, so that their wedge is measured before Proposition 19 reached the ballot.[6] Next, we describe the three components of the wedge and their construction.

[6]The wedge is truncated at zero because a home whose market value has fallen below its factored base is assessed at market value, and inheriting its assessment saves nothing.

The tax rate, $\hat{t}$. Proposition 13 caps the general levy at one percent of assessed value, and voter-approved levy adds to this one-percent base. We divide each parcel's total tax bill on the 2008–2009 roll by its assessed value[7] We bound rates between 0.8 and 2.0 percent, which binds for 0.9 percent of parcels at the bottom and 3.4 percent at the top. The median rate is 1.19 percent. Because the bill can include fixed charges that do not scale with value, this ratio can overstate the rate an heir would pay on a higher assessment.

The inherited assessment, $\hat{A}$. Proposition 13 sets a home's base at its purchase price and lets it rise each year by the inflation factor the State Board of Equalization publishes, never more than two percent. The base resets only at a change in ownership. We start from the 2008–2009 assessed value and roll it forward with the published inflation factors or 2 percent, whichever is lower. If the parcel sold at arm's length after the roll date, the base resets to that sale price and rolls forward from the sale year. Family transfers do not reset the base, because Propositions 58 and 193 excluded parent-child and grandparent-grandchild transfers from reassessment. At the 2019 lien date, 64.5 percent of parcels carry the rolled-forward roll value and 35.5 percent a base reset by a later sale.

The market value, $\hat{p}$. Since unsold houses have no price for the year 2019, we estimate a hedonic model to impute the predicted market value for each parcel. Our hedonic model is estimated on arm's-length, non-distressed California sales from 2016–2017. It regresses the log price on the log of floor area, lot size, bedroom and bathroom counts, and age, with ZIP-code by property-type and sale-month fixed effects. We trim gross outliers, refit, and correct the retransformation from logs to dollars with Duan's smearing factor. Every parcel is then valued at the January 2017 price level and moved to the lien date with a county repeat-sales index. The full model values 91.8 percent of parcels, and the rest use a simple coarser method as a fallback. On 133,231 randomly held-out sales, the model explains 0.82 of the variance in log prices, its median absolute error is 11.1 percent, and 72 percent of predictions fall within 20 percent of the recorded price.

Table 5 reports each term. The median parcel pays 1.19 percent, carries an inherited base of about $293,000, and would sell for about $524,000. The wedge is a potential advantage: it is what the current owner's heir would save, conditional on eligibility for the inheritance exclusion. It is not a tax change anyone has paid. Eligibility is only

[7]Our data provides the tax bill and the assessed value for 2008-2009 years. We use the latest year available for each parcel. A parcel with a missing or unusable ratio receives its county's median.

proxied, by the historical homeowner exemption, and the surviving occupant exclusion protects gaps up to $1 million.

Table 5: Components of the estimated annual tax wedge

	All parcels (January 2019)		Positive wedge (January 2019)		Inherited homes (2018–2019 transfers)	
	Median	Mean	Median	Mean	Median	Mean
Effective tax rate $\hat{t}$ (%)	1.19	1.25	1.20	1.26	1.19	1.26
Assessment base $\hat{A}$ ($000)	293	425	265	359	256	336
Market value $\hat{p}$ ($000)	524	689	533	698	440	544
Valuation gap $\max(\hat{p} - \hat{A}, 0)$ ($000)	172.6	296.2	206.6	339.3	134.5	223.0
Annual wedge $\hat{w}$ ($)	2,109	3,781	2,542	4,332	1,655	2,884
Zero wedge (%)	12.7		0.0		12.4	
No market sale since 2007 (%)	61.8		66.0		63.8	
Historical homeowner exemption (%)	57.2		58.1		56.8	
Gap above the $1 million allowance in 2021 (%)	8.1		9.3		5.5	
Parcels	7,852,730		6,854,091		139,191	

Notes: California single-family and condominium parcels on the 2008–2009 roll. $\hat{w} = \hat{t}\max(\hat{p} - \hat{A}, 0)$ in nominal dollars of the baseline year. All-parcel and positive-wedge columns use the January 2019 lien date. Inherited homes are parcels whose first ownership change after January 2017 was a family transfer (a same-surname interfamily or estate deed) recorded in 2018–2019; their terms are measured at the January 2017 lien date, before Proposition 19 reached the ballot. Medians are taken term by term, so the median wedge is not the product of the median rate and median gap. Allowance share: 2021 gap between projected market value and factored base above $1 million.

5.3 Estimation results

5.3.1 Tax wedge distribution and deadweight loss

Figure 8 shows the distribution of the estimated wedge across California single-family and condominium parcels in 2019. Most homes carry a positive wedge as only 12.7 percent have a zero wedge, mainly homes bought after the 2012 trough whose base is still close to market value. Among homes with a positive wedge, the median annual saving is $2,542 in 2019 dollars, as marked by the figure's reference line. The wedge is a creature of tenure, exactly as Proposition 13 implies. Homes with no market sale since 2007 and a historical homeowner exemption carry a median positive wedge of $3,690.

An outside benchmark helps discipline these numbers. The Legislative Analyst's Office put the annual saving on a long-held Los Angeles home at $3,000 to $4,000 (California Legislative Analyst's Office, 2017). For the 1,078,453 Los Angeles County homes with no market sale since 2007, our median wedge in 2017 is $4,018. The two populations differ, since our definition of long-held is coarser, but the estimates are of the same size.

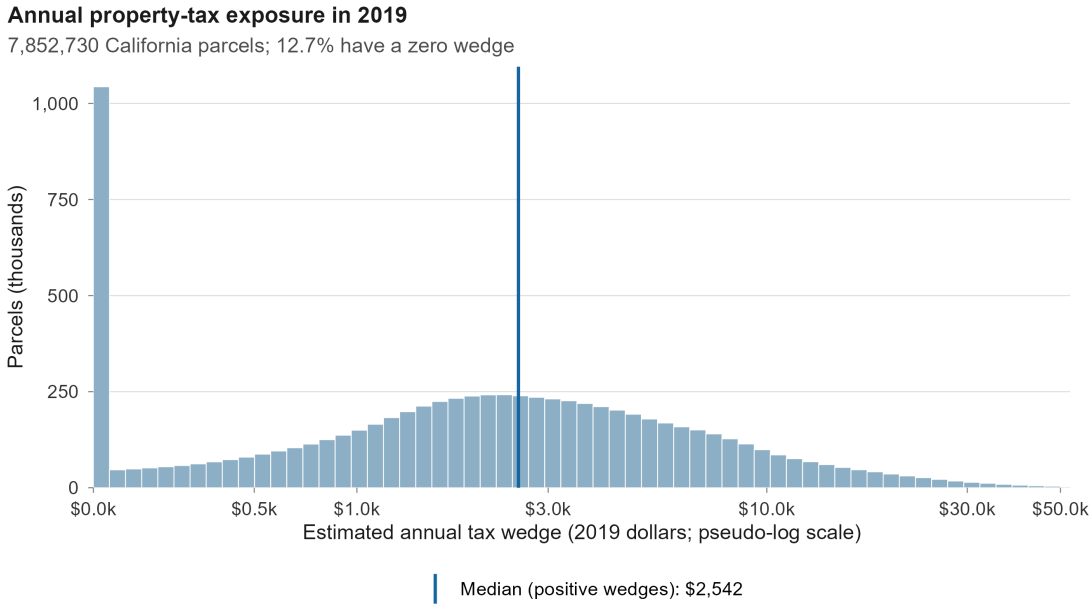


Figure 8: Estimated annual tax wedge $\hat{w} = \hat{t}\max(\hat{p} - \hat{A}, 0)$ for California single-family and condominium parcels at the January 2019 lien date, in 2019 dollars (pseudo-log scale). The reference line marks the median among positive wedges, matching the positive-wedge columns in Table 5. The histogram includes zero wedges.

The homes that matter for welfare are the ones families actually pass down. In 2018–2019, before the reform reached the ballot, 69,596 parcels a year in our universe left their owners through a family transfer (an interfamily deed between parties with the same surname, or an estate deed).[8] Measured at the January 2017 lien date, their median wedge is $1,655, or $2,057 in 2023 dollars, including homes with zero wedges.[9] Together these homes carried about $249 million a year of inherited tax savings. That sum is a transfer from taxpayers to heirs, not a loss to the economy. Most of it goes to heirs who would have kept the home anyway.

The deadweight loss comes only from the heirs the subsidy changes. In the framework, an heir with valuation $u_h \in [u_m - w, u_m)$ keeps a home the market values more. Each

[8]This count includes transfers between spouses, which keep their base under a separate exclusion that Proposition 19 did not change. It misses heirs with a different surname and distributions from trusts. The Legislative Analyst's Office estimated that 60,000 to 80,000 properties a year received the inheritance exclusion (California Legislative Analyst's Office, 2017). The two counts measure different things; they are of the same order of magnitude.

[9]The inherited-home mean is $2,884 in 2017 dollars, or $3,585 in 2023 dollars. It includes zero wedges and refers to the 2018–2019 family-transfer sample. The $4,332 mean in Table 5 instead conditions on positive wedges across the January 2019 parcel stock and is expressed in 2019 dollars.

such home loses $g = u_m - u_h$ per year, somewhere between zero and w. If we make a simplifying assumption that valuations are spread evenly across that interval, the average loss is the Harberger triangle

$$E[g \mid \text{allocation switcher}, w] = \frac{1}{w}\int_0^w g\, dg = \frac{w}{2}. \tag{6}$$

More generally, write the average loss as λw with $0 \leq \lambda \leq 1$; we report $\lambda = 0.25, 0.5, 0.75$. Uniformity is an assumption about a finite reform, not something deadline bunching reveals (Chetty, 2009).

For a median-based illustration, suppose every one of the 27,355 deeds a year that the triple difference finds missing is a home that now goes to a higher-value user. Assigning each home the inherited-home median wedge of $2,057 and setting $\lambda = 1/2$ gives $28.1 million a year of avoided deadweight loss for each annual cohort, in 2023 dollars. This is a representative-home calibration; aggregating heterogeneous wedges instead requires their mean. Using the inherited-home mean gives $49.0 million. Both calculations assume every missing deed reallocates a home. The reallocation share of missing deeds, α, is not identified: a missing deed may be a postponed succession, a title held by a different vehicle, or a family that kept the home for reasons the tax never touched.

6 Conclusion

A large part of American housing turnover never touches a market. From 2007 through 2023, we document that county recorders logged 13.6 million family-transfer deeds, 17 for every 100 market sales. This is a conservative count: it requires a shared surname, so it misses many relatives with different surnames. The ratio ranged between 0.14 and 0.22 across years, near 0.20 to 0.22 through 2012 and falling to 0.14 by 2022–2023. Much of that fall is definitional: the share of interfamily deeds without a surname match, which our count excludes, rose from 17 to 38 percent. We show that this channel is not a recession artifact as it persists through the 2008–2009 recession and beyond, and it is not evenly spread across states. California records 36 family transfers per 100 market sales, about twice the national ratio, and it accounts for 39 percent of the nation's interfamily deeds into or out of trusts. We hypothesize that the tax advantage of inheriting a below-market property-tax assessment along with the house is a key driver of this concentration. We also show that family transfers are more common for middle-aged homes than for new ones, and less common for the most expensive homes. Once a home passes between relatives, it

tends to stay put. Family homes sell slightly faster than market purchases in the first two years, then return to the market more slowly, and three in five homes transferred within families in 2008–2018 had not sold on the open market a decade later. Transaction data built from prices miss this channel entirely.

California's Proposition 19 provides a natural experiment to test whether the channel responds to tax incentives. From June 2020 through the February 2021 deadline, California family deeds ran 21.4 percent above counterfactual, about 25,100 extra transfers, and the deadline month shows the largest seasonally adjusted spike in the series. The 21.4 percent overstates anticipation. California's 2018–2019 base was unusually low, and family deeds were already above counterfactual in early 2020, before the measure reached the ballot. After the deadline, the channel shrank. Net of California's own market-sale slump, a triple difference shows family transfers 22.5 percent lower in 2022–2023 than in 2017–2019, roughly 27,355 fewer deeds a year. A synthetic control and a monthly event study trace the same persistent gap. The early excess deeds equal about 72 percent of the 2021–2022 shortfall, so retiming may explain the first post-deadline years; the 2022–2023 decline is the robust result. Because our count includes spouses, it is broader than the parent-child transfers Proposition 19 targets.

We develop a framework that separates efficient family retention from retention bought by a tax advantage. An heir keeps a home when its value to the family exceeds the market's valuation net of the inherited tax saving, so the saving tips only those heirs whose margin is smaller than the wedge. We measure that wedge parcel by parcel. Homes passed down in California in 2018–2019 carried a median wedge of $2,057 a year in 2023 dollars, including homes with zero wedges. Their total tax savings were about $249 million a year. That sum is a transfer from other taxpayers to heirs, not a loss to the economy. The deadweight loss is the smaller triangle under it. If every missing deed were a home now going to a higher-value user, assigning each home the median wedge and a distortion equal to half that wedge would yield about $28.1 million a year of avoided deadweight loss per annual cohort. This illustration assumes every missing deed reallocates a home. A missing deed may be a postponed succession, a title held through a trust, or a change in legal ownership with no change in who lives in the house.

Three kinds of data would sharpen the answer: longer follow-up on what happens to homes that did not change hands within families, transaction-time records of exclusion claims and assessments, and occupancy and rental status for inherited homes. Together they would show which missing deeds are reallocations and which are delays or relabelings. The policy question is not whether families should keep homes. Many heirs value the

family house more than any buyer would, and the tax code should leave that choice alone. The question is whether inheritance rules in property taxation pay families a premium to keep homes that someone else would value more, and how often that margin binds. Proposition 19 suggests the margin binds often enough to change tens of thousands of recorded family transfers a year in California; how many of those homes would otherwise go to other families remains open. Nevertheless, more evidence is needed to assess how the property tax lock-in prevents the efficient allocation of scarce housing in the United States.

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Appendix

A Data Construction Details

Source and scope. The CoreLogic Owner Transfer file compiles recorded deeds from county recorder offices in all fifty states and the District of Columbia. Our extract contains all record types—deeds of sale, quitclaims, intra-family conveyances, trust transfers, and foreclosure instruments. Coverage density is stable from 2007 onward; recording lags truncate the file around August 2024. All annual statistics use 2007–2023; the Proposition 19 cohort analyses use outcome windows that close by June 30, 2024.

Deduplication. Multi-parcel transactions and re-recorded corrections generate multiple rows per economic event. We keep one record per parcel-date pair, preferring arm's-length category records and, within category, the highest recorded consideration.

Classification. The taxonomy in Section 2 is implemented with the recorder-derived interfamily indicator, the arm's-length category code, recorded consideration, and string features of the buyer and seller name fields (surname equality; exact same-person matches on either buyer slot, which define the same-person retitle class; trust keywords; estate, executor, probate, and administrator keywords). Name fields are populated for the large majority of records (the seller surname is non-missing for 84 percent of interfamily events and the buyer surname for 66 percent); missing names can only push true family transfers into the excluded class of interfamily-flagged deeds without a surname match, never into the market class, but imperfect full-name matching can also misclassify retitling. Missing names are one more reason to read the main definition (same-surname interfamily plus estate deeds) as a conservative count and, in practice, a lower bound on family transfers (Section 2). The 6.1 million excluded deeds enter only the wide definition in Table 2.

Absentee recipients. A recipient is absentee when the buyer mailing ZIP on the deed differs from the property's situs ZIP, among records where both are observed (77 percent of events in 2007, rising to 98 percent by 2023). Time effects absorb common coverage changes; differential changes across states or transfer classes remain a measurement concern.

A.1 Recorded property types

We retain the Owner Transfer land-use and property-indicator codes as text. Detailed labels use five entries from a published CoreLogic crosswalk; remaining codes are retained in the data without an asserted detailed label. The property indicator provides a broader cross-check: in this extract, indicator 11 accompanies both condominium and apartment land-use codes. It therefore cannot distinguish those assets by itself.

Table A1: Recorded property types and family deed flows, 2007–2023

Recorded type	Family deeds	Market deeds	Family share (%)	Family/market
Apartment	7,902	79,892	9.0	0.10
Condominium	866,172	8,670,342	9.1	0.10
Duplex	237,344	1,267,700	15.8	0.19
Single-family	10,596,994	55,480,330	16.0	0.19
Townhouse/rowhouse	209,099	2,258,893	8.5	0.09

Notes: Family deeds are same-surname interfamily and estate/executor deeds (the main definition); trust deeds and interfamily deeds without a surname match are excluded. Market sales carry no price floor. Shares divide family deeds by family plus market deeds. Partial published numeric crosswalk: 163 single-family, 102 townhouse/rowhouse, 112 condominium, 115 duplex, 106 apartment. Other codes remain unverified and are omitted from this exhibit, not the aggregate sample. Types are static recorded characteristics. Apartments are selectively covered: only 13% of source apartment rows pass the existing residential flag. Parcel deeds count neither dwelling units nor occupancy changes.

Types are static recorded characteristics and may differ across records for a parcel. Counts are parcel-date deeds, not dwelling units. In the full source audit, only 13 percent of apartment-coded rows carry the residential flag required by the existing panel. Apartment comparisons therefore do not describe the complete multifamily sector. We preserve the original residential restriction for this reproduction.

Type-specific five-year resale comparisons also differ. Among 2008–2018 cohorts, weighting family and market events to the same market state/cohort distribution gives five-year sale rates of 33.2 versus 27.6 percent for condominiums, 21.3 versus 25.4 percent for duplexes, and 25.3 versus 25.8 percent for single-family properties. Condominiums and townhouses (30.1 versus 27.4 percent) are the exceptions to slower family resale. These comparisons do not identify an effect of inheriting an asset, but they caution against a uniform interpretation of family deeds as lock-in.

The same-surname comparison at ten years also differs by recorded type. Kaplan–Meier estimates leave 66.7 percent of duplex family properties unsold versus 58.8 percent of market purchases, and 59.9 versus 55.9 percent for single-family properties. For condominiums, the ordering reverses: 47.7 percent of same-surname family properties remain

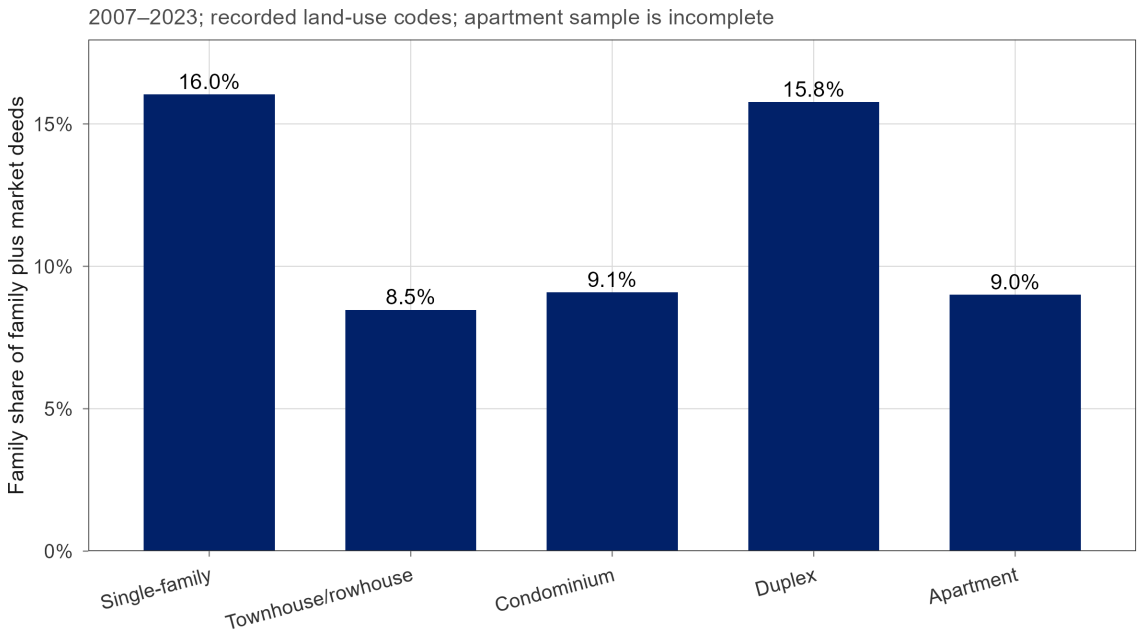


Figure A1: Family share of family-plus-market deeds by recorded type, 2007–2023. Family deeds use the main definition (same-surname interfamily and estate deeds); trust deeds and interfamily deeds without a surname match are excluded. Market sales carry no price floor.

Table A2: Ten-year retention by recorded property type

	Unsold at ten years (%)		Gap
Recorded type	Same-surname family	Market purchase	(pp)
Condominium	47.7	50.4	-2.8
Duplex	66.7	58.8	7.8
Single-family	59.9	55.9	4.0
Townhouse/rowhouse	53.3	51.4	1.9

Notes: Kaplan–Meier survival to the next market sale for 2008–2018 deed cohorts, censored June 2024, using the monthly exit construction of the aggregate analysis. These are pooled within-type comparisons; state/cohort composition, recipient selection, and occupancy histories are not controlled. Apartment results are omitted because of selective coverage.

unsold versus 50.4 percent of market purchases. These pooled comparisons do not hold state and cohort composition fixed.

An exploratory repetition of the transparent 2019-scaled deadline counterfactual yields anticipation excesses of 30.8 percent for single-family properties, 41.4 percent for condominiums, and 70.8 percent for duplexes. These comparisons have neither type-interaction inference nor direct measures of statutory eligibility; they are descriptive heterogeneity checks. Like the aggregate figure, they rest on California's unusually low 2018–2019 base

and overstate anticipation (Section 4). Building type alone does not establish whether the property is a qualifying principal residence.

B Robustness Checks

Trust-inclusive triple difference. Re-estimating the Proposition 19 volume triple difference with all trust deeds added to the family aggregate gives a CA × Post coefficient of −0.165 (s.e. 0.032) and, net of the −0.175 market-sale coefficient, a triple difference of +0.010 log points (state-placebo $p = 0.98$). The decline in direct family deeds disappears in this aggregate. The main reason is weighting. In 2017–2019 California recorded about two trust deeds, three in four of them retitlings into the owner's own trust, for every main-definition family deed. Trust deeds barely fell and rose relative to market sales: their own triple difference is +0.118 ($p = 0.67$). Adding only trust distributions (trust-to-person deeds), the trust deeds most likely to pass a home to heirs, gives a triple difference of −0.232 log points (−20.7 percent; rank 9 of 51, $p = 0.176$), close to the headline. The all-trust aggregate is therefore a weak test of substitution. It cannot rule out that some of the response reflects reclassification into trust vehicles rather than exit from the family channel.

Raw difference-in-differences means. Table A3 reports the unadjusted means underlying the sold-within-24-months estimates.

Table A3: Sold on open market within 24 months: raw cohort means

Cohort group	Region	Pre (2017m1–2018m12)	Post (2021m7–2022m6)	Change
Family transfer	California	15.0% (n=253,737)	11.1% (n=132,599)	−3.9pp
Family transfer	Rest of US	14.1% (n=1,301,954)	11.9% (n=815,529)	−2.2pp
Market purchase	California	6.2% (n=788,934)	4.7% (n=422,157)	−1.5pp
Market purchase	Rest of US	8.3% (n=8,309,691)	8.0% (n=4,853,436)	−0.3pp

Notes: Non-corporate recipients only. Family transfer = same-surname interfamily and estate/executor deeds (same-person retitles and trust deeds excluded); market purchases carry no price floor. Raw difference-in-differences: family −1.7pp, market −1.2pp; raw triple difference −0.5pp. The cell-level regression triple difference is −0.6pp (state-placebo $p = 0.76$).